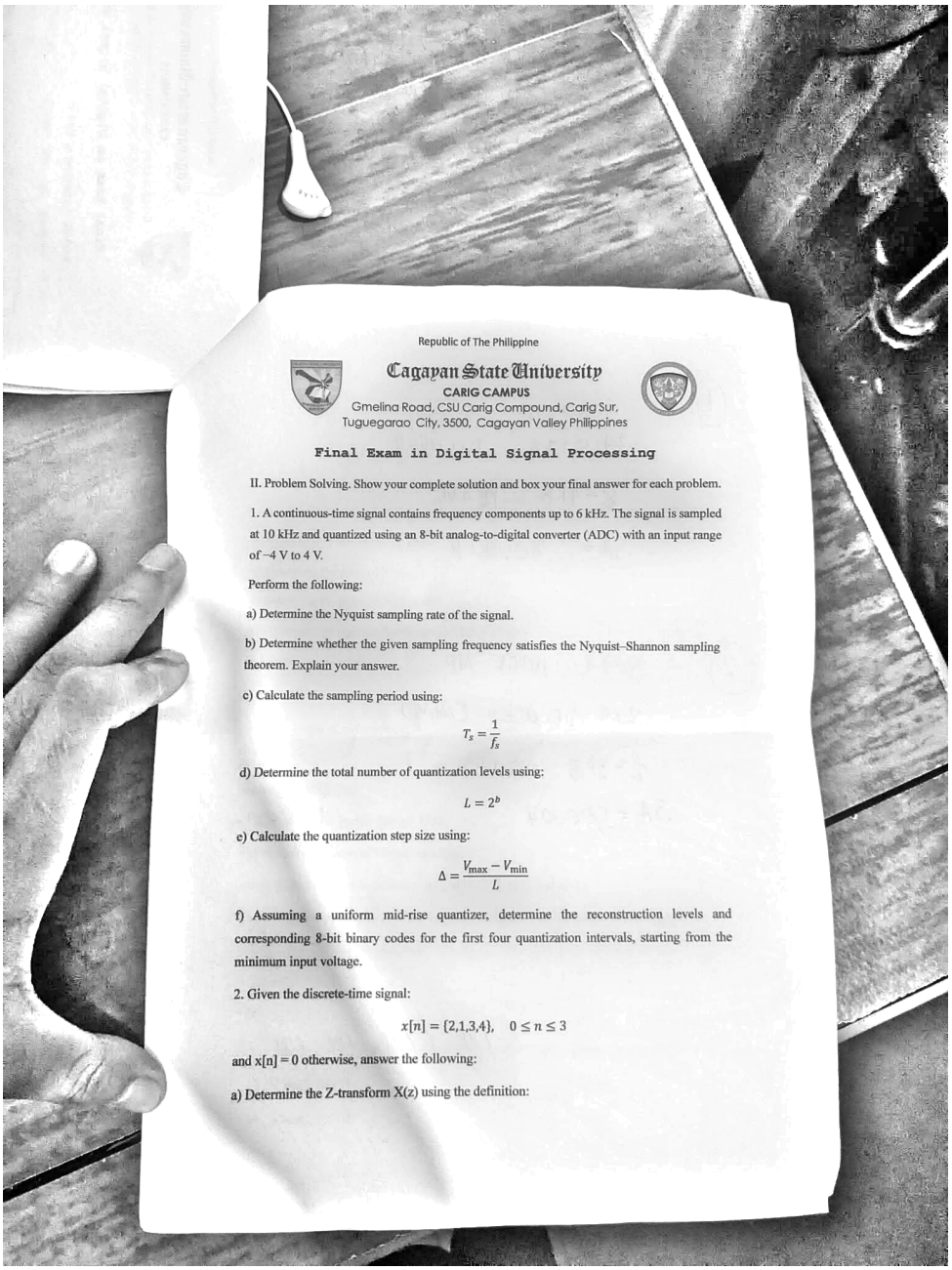
SmartBoard Session Report

Course: cpe14 - DSP
Session: Quantization
Date: 2026-10-24
Status: draft



Republic of The Philippine

Cagayan State University
CARIG CAMPUS
Gmelina Road, CSU Carig Compound, Carig Sur,
Tuguegarao City, 3500, Cagayan Valley Philippines

Final Exam in Digital Signal Processing

II. Problem Solving. Show your complete solution and box your final answer for each problem.

1. A continuous-time signal contains frequency components up to 6 kHz. The signal is sampled at 10 kHz and quantized using an 8-bit analog-to-digital converter (ADC) with an input range of −4 V to 4 V.

Perform the following:

a) Determine the Nyquist sampling rate of the signal.

b) Determine whether the given sampling frequency satisfies the Nyquist–Shannon sampling theorem. Explain your answer.

c) Calculate the sampling period using:

$$T_s = \frac{1}{f_s}$$

d) Determine the total number of quantization levels using:

$$L = 2^b$$

e) Calculate the quantization step size using:

$$\Delta = \frac{V_{max} - V_{min}}{L}$$

f) Assuming a uniform mid-rise quantizer, determine the reconstruction levels and corresponding 8-bit binary codes for the first four quantization intervals, starting from the minimum input voltage.

2. Given the discrete-time signal:

$$x[n] = \{2,1,3,4\}, \quad 0 \le n \le 3$$

and x[n] = 0 otherwise, answer the following:

a) Determine the Z-transform X(z) using the definition:

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